InventOR: Can Large Language Models Set Reorder Points and Safety Stocks? Industrial Evidence Against ERP and Data-Driven Benchmarks

Aris Dressino[a,*]

[a] *TUM School of Management, Technical University of Munich, Arcisstraße 21, Munich, 80333, Germany*

Abstract

Should the reorder point, safety stock and order quantity of a plant-material pair be revised, and can a large language model (LLM) do it? Parameters often remain at ERP values set long ago because revising them takes scarce analyst time. We evaluate an LLM configured only through a prompt, without training, inside a reviewed workflow called *InventOR*: the LLM proposes a reorder-point/order-quantity (r, Q) policy with a written justification, deterministic software gates and replays it on held-out demand, and a planner decides. Using 365 plant-material pairs from three plants of an industrial manufacturer, three LLM runs, seven reference policies (including textbook code that applies the formulas the prompt prescribes) and an ablation withholding the ERP planning parameters, we compare cost, service and inventory in a hard-limit storage replay. The LLM costs about 11% less than ERP-derived parameters, but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines. It reproduces the textbook formulas (median safety-stock and order-quantity ratios of 1.00), with a 9% higher reorder point and about 10% run-to-run variation, whereas code given the same ERP inputs costs 44% more. Once lost sales are

*Corresponding author. Telephone: +49 89 289 25000. ORCID: 0009-0003-0693-9051.
Email address: aris.dressino@tum.de (Aris Dressino)

charged at 1% of unit cost or more, most policies lie within 4% of each other and only a simulation-tuned policy is clearly cheaper. Simulated trajectories exceed storage limits for 53–66% of pairs that pass static checks. The LLM is a feasible, reviewable generator of parameters, not an optimizer; planner benefits remain untested.
Keywords: Inventory Control, Large Language Models, Generative Artificial Intelligence, Safety Stock, Intermittent Demand, Empirical Evaluation

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5, 6], building on the long-standing promise of data analytics in logistics [7].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [8], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that holds when a policy is written down can be violated as stock builds up over time.

Prior LLM work in operations mostly translates problem statements into solver code or

studies agents on synthetic benchmarks (Section 3); we found no evaluation in which an LLM sets per-material (r, Q) parameters for an industrial portfolio and is compared with ERP values and with strong data-driven baselines under controlled information. This paper provides one. We call the workflow *InventOR*: for each plant-material pair, an enterprise LLM application that is configured only through a prompt and receives no project-specific training or fine-tuning proposes a candidate (r, Q) policy together with a justification and an escalation flag. Deterministic software parses, gates and simulates every proposal, and a planner reviews it and alone decides whether any ERP parameter is changed (Section 2).

We study five questions on real industrial data: what cost, service and inventory the LLM policies achieve relative to ERP-derived parameters, to data-driven baselines and to textbook code that applies the formulas the prompt prescribes, when the information available to each approach is controlled (RQ1); why the LLM differs from that code, in terms of formula fidelity, run-to-run noise, demand segments and the price of shortages (RQ2); whether the policies respect storage limits along simulated trajectories (RQ3); how stable they are across independent runs (RQ4); and in which plants and demand regimes a planner should trust, escalate or reject a proposal (RQ5).

We contribute (C1) a workflow with explicit requirements that bound the role of the LLM (Sections 2 and 4); (C2) an information-controlled, trajectory-level evaluation with seven reference arms, including deterministic code that applies the prompt's own formulas, an ablation denied the ERP planning parameters and a hard-limit storage replay (Section 5); (C3) complete-cohort evidence from three LLM runs on 365 plant-material pairs, including the cost decomposition, a break-even analysis of the shortage price and a mechanism analysis of why the LLM does not beat code (Section 6); and (C4) managerial implications and

research perspectives (Sections 7 and 8).

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The LLM reproduces the prompt's textbook formulas almost exactly (median safety-stock and order-quantity ratios to the code arm of 1.00, reorder points 9% higher), so its differences from code are small and noisy, while code that receives the same ERP inputs as the LLM costs 44–45% more than the LLM. When lost sales are charged at 1% of unit cost or more, the ERP-derived, history-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other and only a simulation-tuned policy is clearly cheaper. The simulation therefore does not show superior optimization by the LLM. If only parameters are needed, running the formulas as code is cheaper and exactly repeatable; the LLM adds a written justification, an escalation flag and no coding effort, and whether that helps planners remains to be tested.

2. Problem Setting and Workflow

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. In the classification of Gorry and Scott Morton [9] it is a partly structured operational-control decision: the calculation is textbook, but the choice of estimator, the treatment of intermittent or trending

demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope, and the LLM is a candidate generator that does not forecast, place orders, simulate or decide.

2.2. The InventOR workflow and its level of automation

> **Definition.** *InventOR is a human-reviewed planning workflow that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In terms of the classical decision-support components of data, model and dialog [10, 11], the LLM and the simulator form the model component, the material-scoped data extract and the prompt form the data and knowledge components, and the review interface is the dialog component (Section 4.1). In terms of the levels of automation of Parasuraman et al. [12], InventOR operates at a low-to-intermediate level: it selects and suggests one alternative and leaves execution to the human. Accepting automated suggestions invites complacency and

automation bias [13], which the review rules of Section 4.3 are meant to counter.

2.3. Requirements on the workflow

The requirements follow from the known unreliability of LLM outputs [8], guidance on human–AI interaction and accountability [14, 15], run-to-run nondeterminism of LLM inference [16] and the constraints of inventory control. Table 1 ties each to a design feature (Section 4) and an evaluation (Section 6).

Table 1: Requirements on the workflow, the features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.9
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.8
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

3. Related Work and Positioning

Human–AI decision making in planning.. Research on decision support has long stressed that the value of such a system lies in improving the judgement of a human decision-maker rather than in replacing it [9, 10, 11], and research on human–automation interaction shows that the way suggestions are scoped, presented and verified determines whether users rely on them appropriately or over-rely on them [12, 13]. In inventory control, human override of algorithmic safety-stock advice is a documented risk [17], and model-driven support shapes how groups handle disagreement [18]. Behavioural evidence shows that people discount algorithms after seeing them err [19], and an empirical logistics case finds efficiency advantages

where AI-based route planning is followed in full [20]. InventOR therefore keeps the planner as decision-maker and makes the evidence for each proposal explicit.

LLMs in operations and supply-chain decisions.. Reviews and frameworks map the opportunities for generative AI in supply chains [1, 2, 3, 21, 4, 5, 22] and stress that adoption depends on assigned human roles, data quality and governance [6, 23, 24, 25, 26]. Evidence from manufacturing planning is thin: a fine-tuned LLM for digital-manufacturing optimization [27], a transformer trained to set order timing and quantity and benchmarked against (r, Q) [28], and experiments showing that ChatGPT reproduces human decision biases of managers [29]. Most technical LLM work translates problem statements into optimization models and solver code [30, 31, 32, 33], studies multi-agent control in inventory flows [34, 35], disruption modelling [36] or classical OR tasks such as routing [37]. The closest conceptual neighbour studies how OR algorithms, LLM agents and humans complement each other in multi-period inventory control on a benchmark of more than 1,000 instances, and reports that OR-augmented LLM methods outperform either alone [38]; we do not test OR-augmented generation (Section 8) and ask the prior question of how far an LLM alone gets on industrial data. In forecasting, LLMs do not consistently outperform human judgment in retail [39]. A further LLM study uses an LLM as a forecasting copilot for spare-parts demand, combined with simulation, and reports high variability for irregular demand that can be reduced by averaging repeated queries and enriching prompts [40]. We test run-to-run variability directly (Section 6.8); the difference is that our LLM proposes a complete policy, not a forecast.

Data-driven inventory control without LLMs.. Learning-based replenishment is well established, including data-driven policies learned from sequentially observed non-stationary de-

mand [41] and forecasts augmented with leading indicators [42], and heuristics remain competitive in constrained settings such as perishable inventory [43]. Deep reinforcement learning has been applied to omnichannel and multi-echelon systems [44, 45, 46] and aerospace replenishment [47]; deep-learning forecasting has been integrated with replenishment [48]; contextual distributionally robust optimization uses side information to set order quantities [49]; and distribution-free online algorithms handle capacitated warehouses under uncertain price and demand [50]. The closest operational neighbour is a trained data-driven framework that jointly sets replenishment parameters, safety stock and safety lead time on real automotive data [51]. Reinforcement learning has also been combined with demand forecasting to optimize order points [52]. A longer line of work links forecasting to inventory parameters: safety stock under causal demand forecasts [53], the distribution of combined-forecast errors [54] and forecasting models optimized for inventory cost rather than accuracy [55]. Such methods need training or tuning; InventOR needs none, but its generator cannot be inspected beyond its written justification. Applications published in *Expert Systems with Applications* include model-based and contextual reinforcement learning for retail and supply-chain inventory [56, 57], reinforcement learning for perishable goods [58, 59], transformer and graph-network replenishment planning [60, 61] and time-series mixer models for demand forecasting and inventory [62]; recent work there also couples LLMs with supply-chain decision support [63, 64]. We found among these no evaluation of a prompt-configured LLM that sets per-material (r, Q) parameters and is compared with ERP values and with textbook code under controlled information.

Intermittent and spare-parts demand.. Material histories in manufacturing are frequently intermittent or lumpy. Work on this topic began with the bias of Croston's estimator and the stock-control performance of intermittent-demand estimators [65, 66] and continues with neural [67], bootstrapping [68], obsolescence-aware [69, 70] and state-space [71] forecasts, order-up-to control with intermittent integer demand [72], safety-stock placement [73] and data-driven lead-time forecasting [74], all of which bear on the inputs of the policies we compare. The wider literature offers specialized estimators and classification schemes [75, 76, 77, 78, 79], spare-parts forecasting across life-cycle phases [80] and multi-echelon stocking models for spare parts [81].

InventOR is not a new inventory model and does not claim to beat trained or optimized methods; its (r, Q) policy is the standard workhorse of continuous-review control [82, 83, 84].

4. The InventOR Workflow

4.1. Architecture and components

Following the classical decision-support components [10, 11], InventOR has a *data* component (a material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters and ERP reference values; the fields supplied are controlled and logged; DR4), a *knowledge* component (the system prompt file defining the task, the prohibited methods, the storage rules and the output schema, Supplementary File S1; DR1, DR5), a *model* component split into a generator (an enterprise LLM application with a code interpreter, configured only by the prompt, proposing one (r, Q) policy with a justification and an escalation flag; DR1) and a deterministic verifier and simulator (parser, gate, held-out simulation with hard storage-limit replay and reference arms; DR2, DR3), a *dialog* compo-

nent (policy, justification, flag and simulated comparison shown to the planner under the trust, escalate and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call log of input hash, time, session and raw response, repeated runs; DR5). The generator is the only non-inspectable component. In the evaluated prototype the dialog component is the per-pair output record; no interactive interface was built or evaluated.

4.2. Operation in four steps

For one material, InventOR answers how much safety stock to hold, at what level to reorder and how much to order, in four steps (Figure 1).

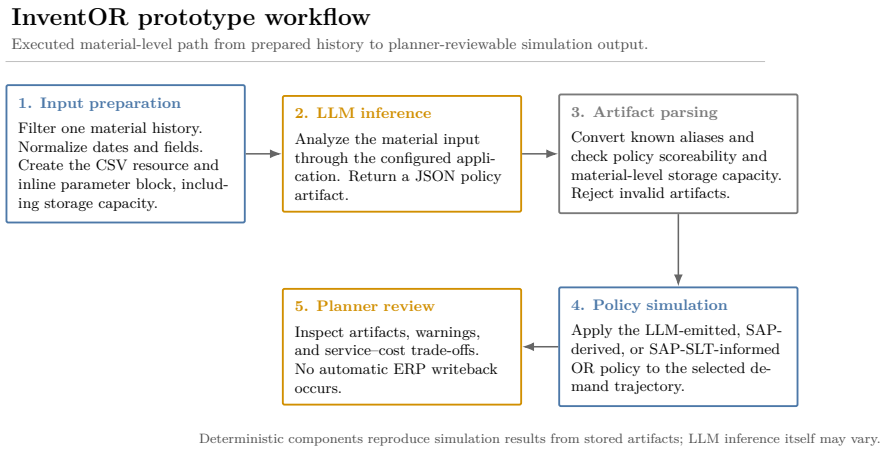


Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

Step 1. Generate. The LLM, accessed through an enterprise application with a code interpreter, receives one material's history (daily demand, realized lead times, working-day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [85] instructs it to compute demand and lead-time statistics, apply an economic-

order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's demand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing [86, 87], together with the comparator policies, producing holding and ordering cost, fill rate and stockout days.

Step 4. Review. A planner sees the policy, the LLM's justification and flag, and the simulated comparison. Only the planner changes ERP parameters. The planner's decision (accept, adjust or reject), the review time and an optional free-text comment are logged with the call, so that practitioner feedback accumulates as a by-product of use; the prototype evaluated here does not yet implement this logging.

The LLM is a *candidate generator* whose output is never trusted without the gate; it does not forecast, place orders or run the simulation. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing [14, 15]. The rules and the 0.5-point threshold are design proposals not evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (apply after review) when the proposal passes the gate, the simulated trajectory respects the storage limit when orders are truncated at it, and the simulated cost is not above that of the best history-only OR reference (Croston/SBA or empirical quantile) with a fill rate not more than about 0.5 points lower.

Escalate when the LLM raises an escalation flag, the material is lumpy or has strongly seasonal or trending demand, the plant or calendar data were repaired, or the policy differs strongly between repeated generations.

Reject when the proposal fails the gate after the retry budget, exceeds the storage limit in the hard-limit replay, or is dominated (higher cost and lower fill) by a history-only reference.

4.4. Provenance and reproducibility of the configuration

The system prompt, a plain-text file deployed unchanged in the enterprise application, defines the task, forbidden methods, capacity rules and output schema (Supplementary File S1). Every call logs the SHA-256 of its input, UTC time, session identifier and raw response. The platform does not expose the model identifier or sampling settings; the evaluated model (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared. Reproducibility rests on the prompt file, the call logs and three independent runs (Section 6.8).

5. Application and Evaluation Design

5.1. Evaluation approach

The evaluation addresses the requirements of Table 1. The primary basis is the hard-limit replay with working-day receipts and no shortage penalty; sensitivity analyses vary these choices. The evaluation is an ex post simulation study: real data and a faithful simulation but no live users. The prompt prescribes a closed-form recipe (an economic order quantity with a minimum-order-quantity floor and a normal-approximation safety stock), so the history-only, SLT-informed and ERP-floor OR arms of Section 6.3 are the deterministic-code counterparts of what the LLM is asked to do, with the same formulas and, depending on the arm, the same inputs. It establishes technical efficacy, feasibility and reliability, not usability, adoption or improved planner decisions.

5.2. Industrial application and data

The data come from a large manufacturer, and the evaluation is retrospective. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the storage limit, so no comparator can satisfy both rules; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date

majority of Plants A and B and test this repair in Section 6.9. Data quality is a known threat to decision support [88] and inventory records [89, 90, 91].

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use.

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability [92], and planned lead times can drift upward [93]. The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Three further arms use no ERP input and represent stronger data-driven practice. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [75, 76] (smoothing constant 0.1), uses the root mean squared one-step forecast error as demand spread and the history-only lead-time moments, which suits intermittent demand [77]. The *empirical-quantile OR* sets the reorder point to the service-level quantile of historical rolling lead-time demand over the rounded mean lead time. The *simulation-tuned* (r, Q) policy selects, from a grid of reserve factors and order-quantity multiples around the history-only policy, the cheapest candidate whose fill rate on the last 250 working days *before the cutoff* meets the service target (or the best-fill candidate if none does). No post-cutoff data are used in any of these arms. We do not

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP inputs used	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock	ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time	OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both	OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None	ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None	Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None	Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None	Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both	Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to "not supplied"	None	Ablation: what the LLM does when it cannot anchor on ERP

implement a deep reinforcement-learning baseline [45, 44]; this is a limitation (Section 8).

5.4. Simulation, capacity replay and metrics

Each pair is simulated independently on the post-cutoff horizon as a single-item, single-echelon continuous-review (r, Q) system [82, 94, 95, 84, 83] with lost sales [96], in the spirit of simulation-based experiments on supply and timing uncertainty [97], a shared opening stock, an empty pipeline and arrivals after the rounded mean lead time in working days (Appendix Appendix A). MOQs are treated as hard order-size constraints, as imposed by suppliers [98], and storage limits as capacity bounds on the replenishment decision [50]. The primary metrics are the mean material-horizon holding-plus-ordering cost, the mean

material-level fill rate, the aggregate (volume-weighted) fill rate and the number of stockout days. We report paired differences with 10,000-sample bootstrap 95% intervals.

A static check $SS + Q \leq$ `max_storage_units` does not show that storage limits hold as inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit on arrival. We therefore record the post-receipt on-hand stock for every policy and day and report the share of pairs and days above the limit (the *unconstrained* simulation), and we *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*), which is our primary evaluation.

5.5. Repeated runs and sensitivities

To examine run-to-run variation, the entire cohort is generated three times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1 serves as the reference for tabulated results unless stated otherwise; runs 2 and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals), the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 6.4) return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm

holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results use the hard-limit replay on the 346-pair common-feasible cohort with working-day receipts and no shortage penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares all arms; its last columns give the LLM cost relative to each arm in the three runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived (98.11% versus 98.30%; interval −0.54 to +0.10). The mean cost difference against SAP-derived is −377 (95% interval −651 to −154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.9 shows that once shortages carry a cost this trade no longer favours the LLM.

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3‡
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8† (vs SAP)

† Other comparisons of the ablation are in Section 6.4. ‡ Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.9).

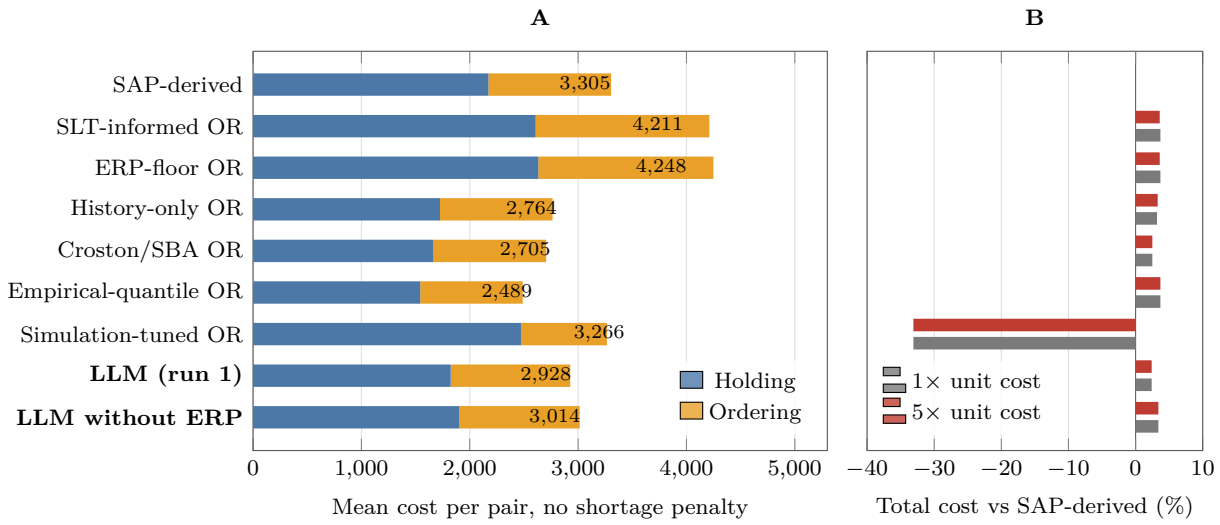


Figure 2: Cost structure and effect of pricing shortages (hard-limit replay, 346 pairs; LLM rows in bold). (A) Mean cost per pair split into holding and ordering cost, with the total at the end of each bar. The LLM and the data-driven ORs carry less inventory than SAP-derived parameters and the ERP-informed ORs. (B) Total cost when lost sales are charged at one and five times the unit cost, relative to SAP-derived parameters (negative: cheaper). Once shortages are priced SAP-derived parameters beat the LLM, and the simulation-tuned policy is cheapest.

inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to

+362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM's advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.9).

Third, the LLM improves on ERP-derived parameters (11.3–11.4% cheaper at 0.14–0.40 points lower fill) but not on the best classical baselines, which also yield a policy for every material at lower cost.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives the same inputs as in runs 1–3, but the ERP safety stock and safety lead time are blanked in the CSV and marked "not supplied"; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters. All 365 pairs returned a valid, storage-compliant policy, and the policies change only slightly beyond run-to-run variation: median ratios to run 1 are 1.00, and the reorder point differs by more than 10% in 54.6% of pairs (48.3% and 48.0% between ERP-supplied runs). In the hard-limit replay the cost is 3,014 at a mean fill of 98.06%, against 2,928–2,931 in runs 1–3. It is 8.8% below SAP-derived parameters (interval −566 to −61; fill −0.24 points, interval including zero) and 28.4–29.1% below the SLT-informed and ERP-floor ORs, but 9.0% above

the history-only OR (interval +134 to +387; cheaper in 91 of 346 pairs), 11.4% above the Croston/SBA OR, 21.1% above the empirical-quantile OR and 7.7% below the simulation-tuned policy. By plant the pattern is unchanged (−10.8% in Plant A, −21.9% in Plant B, +41.6% in Plant C against +36.2% in run 1), so the Plant C loss does not stem from reliance on ERP parameters. The information-asymmetry concern therefore does not explain the LLM's advantage over ERP-derived parameters, and the ERP planning fields appear to be of limited use to the LLM. The ablation withholds only these two fields; unit cost, holding rate, ordering cost, MOQ, service target and storage limit remain available, as they are for all reference arms.

6.5. Storage limits along simulated trajectories

All 365 LLM policies satisfy the static rule $SS + Q \leq$ `max_storage_units`, yet in the unconstrained simulation post-receipt on-hand stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for the SLT-informed OR, because inventory on order and lead-time demand add to stock above $SS + Q$. The static check is thus necessary but not sufficient for any method. Under the hard-limit replay no arm exceeds the limit by more than one unit; enforcing it raises the LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect (Figure 3, hard-limit replay, run 1). The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar

in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.9). Against the data-driven baselines the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR, and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and C. The SLT-informed and history-only ORs coincide in Plant C because the ERP safety lead time is zero for all its pairs.

By demand regime, the 33 lumpy materials show the largest reduction against SAP-derived (−25.4% in run 1) and the 313 smooth materials −9.8% at a 0.24-point lower mean fill. On lumpy materials the service effect is inconsistent (mean fill +0.38 points in run 1, −0.29 in run 2), the history-only and Croston/SBA ORs cost about 9–10% less than the LLM, and only the empirical-quantile OR is slightly more expensive (1.2%); with 33 materials this is indicative only. By terciles of the coefficient of variation, the LLM costs 15.7%, 7.5% and 12.9% less than SAP-derived (lowest to highest), with a 0.68-point lower mean fill in the middle tercile.

6.7. Why the LLM does not beat code

The prompt prescribes an economic order quantity with a minimum-order-quantity floor and a normal-approximation safety stock (Appendix Appendix A). The History-only OR applies these formulas as code to the same demand and lead-time history, and the SLT-informed and ERP-floor ORs apply them with the ERP inputs the LLM also receives (Section 5). The comparison with code is therefore direct.

The LLM reproduces the formulas.. Pair by pair, the median ratio of the LLM order quantity to the History-only order quantity is 1.00 (medians of 2,810 and 2,828 units) and that of the

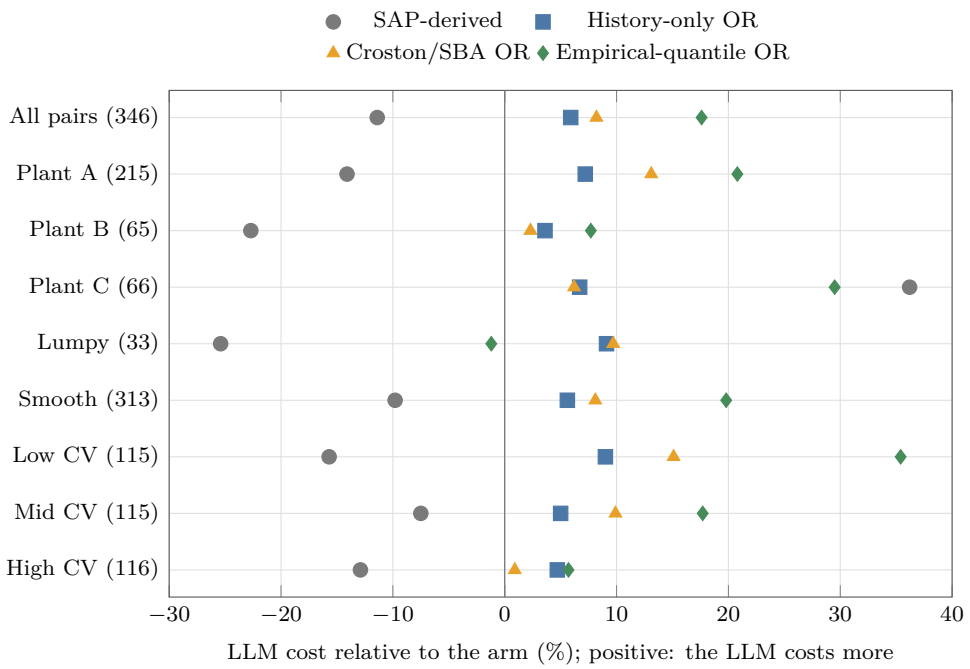


Figure 3: Where the LLM wins and loses (run 1, hard-limit replay). Each marker is the LLM's mean cost relative to one comparator in a segment of pairs (number of pairs in brackets); markers to the right of the vertical line mean the LLM costs more. The LLM is cheaper than SAP-derived parameters in Plants A and B but not in Plant C, and costs more than the three data-driven baselines in almost every segment.

safety stock is 1.00 (712 and 718 units, against an ERP median of 0). The reorder point is 9% higher than the code's in the median pair, in every segment of Table 4, and the LLM costs 4–9% more than the History-only code in every segment. The difference to code is thus a small, systematic uplift of the reorder point rather than a different policy.

Inputs, not the LLM, decide the ranking against code.. Code that receives the same ERP inputs as the LLM costs 4,211 (SLT-informed OR) and 4,248 (ERP-floor OR) per pair against 2,928 for the LLM in run 1, i.e. 44–45% more than the LLM, because the ERP safety lead times inflate the textbook reorder point. Code that ignores the ERP planning inputs costs 2,764, 5.9% less than the LLM. The LLM resembles the History-only code rather than the ERP-inflated code, which is consistent with the ablation in Section 6.4.

Run-to-run noise.. The cost of a pair varies by 10.5% of its mean between the three runs, and the runs disagree on whether the LLM beats the History-only code for 34.1% of the pairs and on whether it beats SAP-derived parameters for 18.8%. The LLM mean is cheaper than the History-only code for 17.1% of the pairs and cheaper than SAP-derived parameters for 76.3%. Taking the best of the three runs for each pair, which is only possible with hindsight, gives a mean cost of 2,765, equal to that of the History-only code (2,764); the worst of three gives 3,146. Repeatability, which code has by construction, is therefore a practical cost of using an LLM here.

Plant C and segments.. Table 4 shows that the LLM advantage over SAP-derived parameters disappears in storage-bound pairs (+4.9% against -14.9% elsewhere) and reverses in Plant C (+33.3%); it does not vary monotonically with demand variability. Plant C is not an LLM-specific failure: the History-only code is also +28% more costly than SAP-derived parameters there, so the ERP parameters are leaner than any formula-based policy in that plant when lost sales are not charged. Plant C also has the least regular demand (median coefficient of variation of daily demand 0.61, average inter-demand interval 1.25 days and lead-time coefficient of variation 0.78, against 0.51, 1.04 and 0.63 over all pairs). In the 188 pairs with a zero ERP safety stock the LLM is 8.4% cheaper than SAP but 7.5% costlier than the code; with no shortage price, the code's extra safety stock earns nothing in the primary basis.

Table 4: Where the LLM differs from code. Mean cost per pair in the hard-limit replay with zero shortage penalty (LLM: mean of three runs; History-only: textbook code on demand and lead-time history; SAP: ERP-derived), the LLM cost relative to each (in %), and the median ratio of the LLM reorder point to the textbook reorder point. Storage-bound pairs are those for which the textbook safety stock plus order quantity exceeds the storage limit, so that the textbook policy is projected down to it.

Segment	n	LLM	History-only	SAP	vs History-only (%)	vs SAP (%)	ROP ratio
All pairs	346	2,929	2,764	3,305	+6.0	-11.4	1.09
Plant A	215	2,370	2,231	2,782	+6.2	-14.8	1.08
Plant B	65	5,050	4,746	6,360	+6.4	-20.6	1.10
Plant C	66	2,662	2,549	1,997	+4.4	+33.3	1.11
Demand CV, lowest third	115	2,249	2,092	2,707	+7.5	-16.9	1.09
Demand CV, middle third	115	3,872	3,659	4,152	+5.8	-6.8	1.10
Demand CV, highest third	116	2,669	2,544	3,057	+4.9	-12.7	1.08
ERP safety stock zero	188	2,946	2,741	3,215	+7.5	-8.4	1.09
ERP safety stock positive	158	2,909	2,792	3,411	+4.2	-14.7	1.09
Storage-bound	145	1,491	1,408	1,422	+5.9	+4.9	1.10
Not storage-bound	201	3,967	3,743	4,663	+6.0	-14.9	1.08

6.8. Run-to-run stability

Across the three runs the aggregate results are very similar (Figure 4): the LLM cost relative to SAP-derived is −11.4%, −11.3% and −11.4%, and the median reorder points are close (1,584, 1,582 and 1,466). Individual policies vary, however. Between runs 2 and 3, all three values are identical for only 36 of 365 pairs and the reorder point differs by more than 10% in about 48% of pairs. Aggregate behaviour is reproducible, but a single generation per material is a noisy estimate that should be reviewed [16].

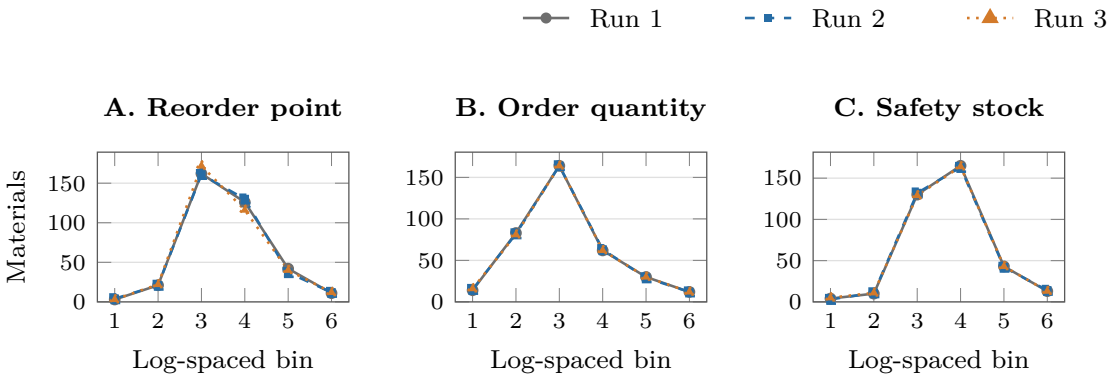


Figure 4: Distribution of the proposed reorder point (A), order quantity (B) and safety stock (C) over the 365 materials in the three runs, on six common log-spaced bins (reorder point 3–683,000; order quantity 63–722,000; safety stock 0–289,000 units). Nearly coinciding lines show reproducible aggregate behaviour.

6.9. Sensitivity analyses

Plant C calendar.. Plant C's working-day flag required reconstruction, so we re-scored all policies under four alternatives (any-plant, strict-all, exclusion of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM remains 31–35% more expensive than SAP-derived there in the three full runs, and the cohort-wide advantage is −10.8% to −11.1%; the strict and exclusion variants (280 pairs) show −17.0% to −17.6%. The Plant C result is therefore not an artefact of the repair, although the LLM saw demand series built on the repaired calendar.

Shortage penalty.. Figure 2B gives mean total cost per pair (holding, ordering and shortage cost) relative to SAP-derived parameters in the hard-limit replay for penalties of one and five times the unit cost. Costs are dominated by the shortage term, and the differences among the non-tuned arms (0.1–3.7%) should be read with caution. The simulation-tuned policy is by far the cheapest (1,094,043 at one times the unit cost, about 33% below SAP-derived), because it was selected to meet the service target on the pre-cutoff history. Among the other arms SAP-derived parameters are the cheapest, as they carry more inventory and fewer stockouts. LLM run 1 then costs 2.4% more than SAP-derived at both multiples, is about as expensive as the Croston/SBA OR (1,673,003 versus 1,674,925 at one times), and is cheaper than the SLT-informed (1.2%), ERP-floor, empirical-quantile and history-only (0.8%) ORs. The LLM's cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged.

Break-even shortage price.. Table 5 re-scores the same replays for a lost-sales penalty between zero and one unit cost per unit. The LLM is cheaper than SAP-derived parameters only

while the penalty stays below 0.9% of the unit cost, and cheaper than the History-only and empirical-quantile code only above 1.5% and 2.4%, by about 1% or less. It is cheaper than the SLT-informed and ERP-floor ORs at every penalty and cheaper than the simulation-tuned policy only below 0.06%. From a penalty of 1% of the unit cost, the SAP-derived, History-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other, while the simulation-tuned policy is 28–33% cheaper than SAP-derived parameters. The ranking between the LLM and the formulas is thus decided by a shortage price that the primary basis sets to zero, which is why it is reported as a sensitivity and not as a headline result.

Table 5: Break-even shortage price. Mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay when each lost unit is charged at the given multiple of the unit cost, and the break-even multiple at which the LLM and the arm have equal mean cost (common-feasible cohort, $n = 346$).

Arm	0	0.01	0.1	1	Break-even multiple of unit cost
SAP-derived	3,305	19,613	166,388	1,634,136	0.0089; LLM cheaper below
History-only OR	2,764	19,609	171,207	1,687,191	0.0145; LLM cheaper above
Croston/SBA OR	2,705	19,427	169,927	1,674,925	none; LLM costlier on the grid
Empirical-quantile OR	2,489	19,402	171,618	1,693,784	0.0242; LLM cheaper above
Simulation-tuned	3,266	14,174	112,344	1,094,043	0.0006; LLM cheaper below
LLM (mean of three runs)	2,929	19,660	170,238	1,676,014	–

7. Managerial Insights and Best Practices

1. **Position the tool as a reviewed planning aid, not an autonomous engine.** Single generations differ at the material level (Section 6.8). The gate-simulate-review structure yields a justified candidate for every material and leaves the decision to the planner; whether this helps planners is untested.

2. **Benchmark against data-driven baselines.** ERP-derived parameters and ERP-informed ORs are weak benchmarks (safety stocks are often stale or zero; safety lead times inflate buffers), so comparing only with ERP overestimates the benefit: the LLM

cost 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines.

3. **Expect plant-level differences.** Against ERP-derived parameters the LLM was cheaper in Plants A and B (14–23%) and on 33 lumpy materials (about 25%, mixed service) but not in Plant C; we did not test whether escalation rules would catch such cases.

4. **Verify storage along the trajectory.** The static rule $SS + Q \leq$ limit cannot protect a warehouse: 53–66% of pairs breach the limit in simulation under each of the three arms checked. Enforce a cap in the order-release logic or treat the limit as a design requirement.

5. **Price shortages explicitly.** Without a shortage penalty the ranking rewards understocking; with a penalty of one unit cost the LLM is about 2.4% more expensive than SAP-derived. Set service targets and penalties before reading cost comparisons.

6. **Freeze and log the configuration.** Treat the system prompt as a controlled file, log every call (input hash, time, session, raw response) and re-run before changing the prompt.

7. **Scale in stages.** Automate generation and gating first, then improve data quality, refine the prompt under freeze-and-log, and widen the scope; re-run the evaluation after every change.

8. **Cost and latency.** Each generation took several minutes, whereas OR baselines take milliseconds; periodic regeneration would support adaptable planning [99].

8. Discussion, Limitations and Research Perspectives

What the evidence shows.. A prompt-configured LLM generates feasible, reviewable (r, Q) policies for every material; in simulation they cost less than ERP-derived parameters but more than data-driven OR policies. InventOR is therefore not presented as outperforming optimization: its role is a reviewed candidate generator, and whether that helps planners is a hypothesis the simulation cannot test. The mechanism analysis (Section 6.7) explains why: the LLM applies the textbook recipe it is given, with a small reorder-point uplift and run-to-run noise, so it cannot beat the same formulas run as code on the same inputs. For an organization that only needs parameters, running those formulas, or a tuned policy once shortages are priced, is the cheaper and exactly repeatable choice; the LLM is justified where a written rationale, an escalation flag and the absence of coding effort matter more than a few percent of holding and ordering cost.

Practical principles for LLM-assisted replenishment.. We derive six principles from this application and offer them as hypotheses. **DP1, bound the generator:** the LLM is a candidate generator whose prompt fixes the method, constraints and output schema, with no authority to act. **DP2, verify deterministically and along time:** check outputs with code and feasibility along the trajectory, because point checks can pass while constraints are violated over time (Section 6.5). **DP3, show counterfactual evidence:** give the reviewer simulated comparisons with credible references. **DP4, control information in evaluation:** compare against baselines with and without the same information and withhold incumbent inputs in an ablation (Sections 6.3, 6.4). **DP5, freeze and log:** treat the prompt as a controlled file, log every call and repeat runs (Section 6.8). **DP6, escalate by rule:**

define in advance when proposals are trusted, escalated or rejected, to limit complacency [13] (Section 4.3).

Implementation experience and roadmap.. Informal feedback from the planner responsible for the materials was positive. This single, unstructured report is not a validation and is not evidence of benefit. The planned expansion has four stages, each changing the configuration and therefore requiring the evaluation to be repeated: (1) *automation*, connecting the generator to a workflow-automation tool and the enterprise agent platform so that generation, gating and simulation run without manual uploads (planned for Q4 2026–Q1 2027); (2) *data optimization*, improving data quality (calendars, stale ERP parameters) and integrating further data; (3) *prompt optimization*, refining the prompt for more consistent outputs (Section 6.8); and (4) *scaling and integration* across use cases and the system landscape.

Limitations.. (i) The data come from one manufacturer and three plants. (ii) The model identifier and inference settings are operator-declared. (iii) No planner outcomes were measured (informal planner feedback aside): the benefits of reviewability, auditability and planner support and the rules of Section 4.3 are design intentions that a field study must test. (iv) The simulation uses deterministic arrival at the rounded mean lead time, ignores observed receipts and starts with an empty pipeline, and does not simulate lead-time variability, which may bias comparisons that depend on safety lead time; we did not report service-target attainment. (v) The primary results use a zero shortage penalty; with a positive penalty the ranking changes. Costs are dominated by a few high-demand materials and the 346 pairs are not independent, so the bootstrap intervals understate uncertainty. (vi) The (r, Q) family is single-item and single-echelon, and individual policies vary across runs. (vii) We included no

deep reinforcement-learning or neural forecasting baseline, and the ERP-derived arm (ERP safety stock with our own order quantity) is not an observed record of practice.

Research perspectives..

1. *Ensembling and hybrids.* Aggregating several generations may reduce individual-policy noise [40], and combining the best data-driven OR policy with LLM review or regime-specific choice [100] could capture the OR cost level and the LLM's flags.

2. *Comparison with trained methods.* Benchmarks against deep reinforcement-learning and other trained policies [51, 44, 45] on identical data and information, and richer policy families such as (s, S) [78, 79], would clarify the trade-off between setup effort and policy quality.

3. *Planner-in-the-loop field trials.* The decision log of the review step provides the data for such studies. Prospective studies should compare planners with and without the aid on matched materials, measuring planner time, override rates and realized service and inventory, and should examine adoption drivers [24, 23, 6].

9. Conclusion

We evaluated whether a prompt-configured LLM can set replenishment parameters, a recurring production-planning decision, for 365 plant-material pairs of an industrial manufacturer. Without training, the LLM generated feasible, justified (r, Q) policies for every material, and a deterministic gate, a hard-limit simulation and explicit review rules made them checkable. The proposals cost about 11% less than ERP-derived parameters at slightly

lower service, but the LLM costs about 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies; when lost sales are priced at one unit cost or more, ERP-derived parameters are the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest overall. Storage limits that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The LLM reproduces the textbook formulas it is given, so the same formulas run as code are cheaper and exactly repeatable, and the ranking against code is decided by a shortage price that the primary basis sets to zero. The evidence supports InventOR as a checkable candidate generator inside a reviewed process, not as an optimizer that beats data-driven baselines; the principles of Section 8 are hypotheses, and field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation.. These are the standard textbook calculations [84, 94]. For pair i, let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\mathrm{MOQ}_i, \lceil\sqrt{2K_i \cdot 250\bar{d}_i/h_i}\rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma^2_{D,i} + \bar{d}_i^2\sigma^2_{L,i}}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i}\rceil$ and the reorder point $R_i = \lceil \bar{d}_i L_i^{\mathrm{eff}} + SS_i\rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i. All policies are

projected to $SS_i + Q_i \leq \texttt{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation.. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i \bar{I}_i T/250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\texttt{max_storage_units}_i$ minus the inventory position, and no order is placed if the truncated quantity is zero.

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The raw operational data and material-level artifacts cannot be publicly released because they contain commercially sensitive material, supplier and planning information. A confidentiality-screened release package containing analysis code, the anonymized variable schema, aggregate evaluation results, manuscript sources and prompt files is available at `github.com/DressPD/inventor_public`. A versioned Zenodo archive of the release is available at `https://doi.org/10.5281/zenodo.22262052`. Supplementary File S1 contains the system prompt used for all runs; the prompt file, the runner code and, for each call, a hash of the input, the UTC time, the session identifier and the raw response are in the release. The model identifier and inference settings are not exposed by the platform and are operator-declared (Anthropic Claude Sonnet 4.6).

Acknowledgments

The author thanks the industrial collaborators and material-control experts who contributed domain knowledge and access to anonymized operational data, and the enterprise data and IT teams for supporting data access and prototype execution. Any errors or omissions remain the responsibility of the author.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used OpenAI GPT-5.6 and Anthropic Claude (Sonnet 5.5), accessed through the OpenCode coding assistant, in order to edit language, restructure text, check consistency, verify citations against Crossref metadata,

proof the LaTeX source, and write and run the analysis scripts that produce the reported evidence. These assistants are distinct from the Claude Sonnet 4.6 deployment evaluated in the study. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The figures are data visualizations drawn by scripts directly from the reported evidence; no generative AI created or altered any image.

References

[1] I. Jackson, D. Ivanov, A. Dolgui, J. Namdar, Generative artificial intelligence in supply chain and operations management: A capability-based framework for analysis and implementation, International Journal of Production Research 62 (17) (2024) 6120–6145. doi:10.1080/00207543.2024.2309309.

[2] S. Fosso Wamba, C. Guthrie, M. M. Queiroz, S. Minner, ChatGPT and generative artificial intelligence: An exploratory study of key benefits and challenges in operations and supply chain management, International Journal of Production Research 62 (16) (2024) 5676–5696. doi:10.1080/00207543.2023.2294116.

[3] S. Fosso Wamba, M. M. Queiroz, C. J. Chiappetta Jabbour, C. Shi, Are both generative AI and ChatGPT game changers for 21st-century operations and supply chain excellence?, International Journal of Production Economics 265 (2023) 109015. doi:10.1016/j.ijpe.2023.109015.

[4] Z. Song, Y. Xie, L. Yang, Y. Zhao, Large language models in supply chain management: A systematic literature review and application framework, International Journal of Production Research 64 (16) (2026) 7144–7184. doi:10.1080/00207543.2026.2641103.

[5] S. Panja, T. Zheng, C. H. Glock, Potentials of generative AI in logistics and supply chain management: A guiding review on applications and future research directions, International Journal of Production ResearchAdvance online publication (2026). doi:10.1080/00207543.2026.2680586.

[6] N. Trifone, F. Dallari, A. Brintrup, Drivers and constraints of agentic AI in supply chain management: An exploratory study, International Journal of Production ResearchAdvance online publication (2026). doi:10.1080/00207543.2026.2669878.

[7] G. Wang, A. Gunasekaran, E. W. T. Ngai, T. Papadopoulos, Big data analytics in logistics and supply chain management: Certain investigations for research and applications, International Journal of Production Economics 176 (2016) 98–110. doi:10.1016/j.ijpe.2016.03.014.

[8] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Computing Surveys 55 (12) (2023) 1–38. doi:10.1145/3571730.

[9] G. A. Gorry, M. S. Scott Morton, A framework for management information systems, Sloan Management Review 13 (1) (1971) 55–70.

[10] R. H. Sprague, A framework for the development of decision support systems, MIS Quarterly 4 (4) (1980) 1–26. doi:10.2307/248957.

[11] J. P. Shim, M. Warkentin, J. F. Courtney, D. J. Power, R. Sharda, C. Carlsson, Past, present, and future of decision support technology, Decision Support Systems 33 (2) (2002) 111–126. doi:10.1016/S0167-9236(01)00139-7.

[12] R. Parasuraman, T. B. Sheridan, C. D. Wickens, A model for types and levels of human interaction with automation, IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans 30 (3) (2000) 286–297. doi:10.1109/3468.844354.

[13] R. Parasuraman, D. H. Manzey, Complacency and bias in human use of automation: An attentional integration, Human Factors 52 (3) (2010) 381–410. doi:10.1177/0018720810376055.

[14] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, et al., Guidelines for human-AI interaction, in: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 2019, pp. 1–13. doi:10.1145/3290605.3300233.

[15] I. D. Raji, A. Smart, R. N. White, M. Mitchell, T. Gebru, B. Hutchinson, J. Smith-Loud, D. Theron, P. Barnes, Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing, in: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, 2020, pp. 33–44. doi:10.1145/3351095.3372873.

[16] S. Ouyang, J. M. Zhang, M. Harman, M. Wang, An empirical study of the non-determinism of ChatGPT in code generation, ACM Transactions on Software Engineering and Methodology 34 (2) (2025) 1–28. doi:10.1145/3697010.

[17] X. Wang, V. S. Rodrigues, E. Demir, J. Sarkis, Algorithm aversion during disruptions: The case of safety stock, International Journal of Production Economics 278 (2024) 109442. doi:10.1016/j.ijpe.2024.109442.

[18] A. Engin, L. A. Franco, E. A. J. A. Rouwette, How groups manage conflict when using model-driven decision support: An epistemic motivation lens, Omega 129 (2024) 103142. doi:10.1016/j.omega.2024.103142.

[19] B. J. Dietvorst, J. P. Simmons, C. Massey, Algorithm aversion: People erroneously avoid algorithms after seeing them err, Journal of Experimental Psychology: General 144 (1) (2015) 114–126. doi:10.1037/xge0000033.

[20] D. Loske, M. Klumpp, Human-AI collaboration in route planning: An empirical efficiency-based analysis in retail logistics, International Journal of Production Economics 241 (2021) 108236. doi:10.1016/j.ijpe.2021.108236.

[21] L. Li, Y. Liu, Y. Jin, T. C. E. Cheng, Q. Zhang, Generative AI-enabled supply chain management: The critical role of coordination and dynamism, International Journal of Production Economics 277 (2024) 109388. doi:10.1016/j.ijpe.2024.109388.

[22] T. Boone, B. Fahimnia, R. Ganeshan, D. M. Herold, N. R. Sanders, Generative AI: Opportunities, challenges, and research directions for supply chain resilience, Transportation Research Part E: Logistics and Transportation Review 199 (2025) 104135. doi:10.1016/j.tre.2025.104135.

[23] V. G. Cannas, M. P. Ciano, M. Saltalamacchia, R. Secchi, Artificial intelligence in supply chain and operations management: A multiple case study research, International Journal of Production Research 62 (9) (2024) 3333–3360. doi:10.1080/00207543.2023.2232050.

[24] V. Venkatesh, R. Raman, F. Cruz-Jesus, AI and emerging technology adoption: A research agenda for operations management, International Journal of Production Research 62 (15) (2024) 5367–5377. doi:10.1080/00207543.2023.2192309.

[25] M. C. Cohen, T. Dai, G. Perakis, N. Agrawal, G. Allon, R. N. Boute, G. P. Cachon, Z. Chen, M. Cohen, R. Cristian, et al., OM Forum—Supply Chain Management in the AI Era: A vision statement from the operations management community, Manufacturing & Service Operations Management 28 (3) (2026) 687–705. doi:10.1287/msom.2025.1065.

[26] F. Olan, K. Spanaki, W. Ahmed, G. Zhao, Enabling explainable artificial intelligence capabilities in supply chain decision support making, Production Planning & Control 36 (6) (2025) 808–819. doi:10.1080/09537287.2024.2313514.

[27] P. Amarasinghe, S. Nguyen, Y. Sun, S. Arisian, D. Alahakoon, Business optimization for digital manufacturing: A fine-tuned large language model approach, International Journal of Production Economics 295 (2026) 109934. doi:10.1016/j.ijpe.2026.109934.

[28] M. Liu, Y. Bai, M. Qi, Z.-J. M. Shen, Inventory management with transformer: Automated decision making for order timing and quantity, Service Science 18 (3) (2026) 196–211. doi:10.1287/serv.2024.0236.

[29] Y. Chen, S. N. Kirshner, A. Ovchinnikov, M. Andiappan, T. Jenkin, A manager and an AI walk into a bar: Does ChatGPT make biased decisions like we do?, Manufacturing & Service Operations Management 27 (2) (2025) 354–368. doi:10.1287/msom.2023.0279.

[30] B. Li, K. Mellou, B. Zhang, J. Pathuri, I. Menache, Large language models for supply chain optimization, arXiv preprint arXiv:2307.03875 (2023). doi:10.48550/arXiv.2307.03875.

[31] C. Huang, Z. Tang, S. Hu, R. Jiang, X. Zheng, D. Ge, B. Wang, Z. Wang, ORLM: A customizable framework in training large models for automated optimization modeling, Operations Research 73 (6) (2025) 2986–3009. doi:10.1287/opre.2024.1233.

[32] A. Ahmaditeshnizi, W. Gao, M. Udell, OptiMUS: Scalable optimization modeling with (MI)LP solvers and large language models, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 577–596.
URL `https://proceedings.mlr.press/v235/ahmaditeshnizi24a.html`

[33] B. Zhang, P. Luo, G. Yang, B.-H. Soong, C. Yuen, OR-LLM-Agent: Automating modeling and solving of operations research optimization problems with reasoning LLM, arXiv preprint arXiv:2503.10009 (2025). doi:10.48550/arXiv.2503.10009.

[34] Y. Quan, Z. Liu, InvAgent: A large language model based multi-agent system for inventory management in supply chains, arXiv preprint arXiv:2407.11384 (2024). doi:10.48550/arXiv.2407.11384.

[35] R. Jannelli, M. Schöpf, L. Bickel, T. H. Netland, A. Brintrup, Agentic LLMs in the supply chain: Towards autonomous multi-agent consensus-seeking, International Journal of Production ResearchAdvance online publication (2025). doi:10.1080/00207543.2025.2604311.

[36] K.-Y. Lin, S.-Y. Wu, K. Matsuno, Generative AI-driven resilience in supply chain management: UNISONE framework for disruption modelling and capacity optimisation, International Journal of Production ResearchAdvance online publication (2025). doi:10.1080/00207543.2025.2574412.

[37] S. Hamdan, N. Brahimi, Evaluating large language models for complex supply chain optimisation: A focus on routing problems, International Journal of Production ResearchAdvance online publication (2026). doi:10.1080/00207543.2026.2737946.

[38] J. Baek, Y. Fu, W. Ma, T. Peng, AI agents for inventory control: Human-LLM-OR complementarity, arXiv preprint arXiv:2602.12631 (2026). doi:10.48550/arXiv.2602.12631.

[39] M. Abolghasemi, O. Ganbold, K. Rotaru, Humans vs. large language models: Judgmental forecasting in an era of advanced AI, International Journal of Forecasting 41 (2) (2025) 631–648. doi:10.1016/j.ijforecast.2024.07.003.

[40] M. Abbaszadeh Nakhost, A. Bloemer, S. Minner, Full automation or just a copilot? a study of large language models on spare parts demand forecasting with OpenAI o1-mini, International Journal of Production ResearchAdvance online publication (2025). doi:10.1080/00207543.2025.2591225.

[41] K. Ren, H. Bidkhori, Z.-J. M. Shen, Data-driven inventory policy: Learning from sequentially observed non-stationary data, Omega 123 (2024) 102942. doi:10.1016/j.omega.2023.102942.

[42] Y. R. Sagaert, N. Kourentzes, Inventory management with leading indicator augmented hierarchical forecasts, Omega 136 (2025) 103335. doi:10.1016/j.omega.2025.103335.

[43] S. Gulecyuz, B. O'Sullivan, S. Armagan Tarim, A heuristic method for perishable inventory management under non-stationary demand, Omega 133 (2025) 103267. doi:10.1016/j.omega.2024.103267.

[44] M. Kolyaei, L. Zhang, M. Blom, Inventory replenishment and fulfilment decisions for an omnichannel retailer: A reinforcement learning-based method, International Journal of Production Research 63 (24) (2025) 9571–9592. doi:10.1080/00207543.2025.2520596.

[45] F. Stranieri, F. Stella, C. Kouki, Performance of deep reinforcement learning algorithms in two-echelon inventory control systems, International Journal of Production Research 62 (17) (2024) 6211–6226. doi:10.1080/00207543.2024.2311180.

[46] S. Ma, Omni-channel retail inventory management with deep reinforcement learning, Omega 144 (2026) 103586. doi:10.1016/j.omega.2026.103586.

[47] H. Wang, J. Tao, T. Peng, A. Brintrup, E. E. Kosasih, Y. Lu, R. Tang, L. Hu, Dynamic inventory replenishment strategy for aerospace manufacturing supply chain: Combining reinforcement learning and multi-agent simulation, International Journal of Production Research 60 (13) (2022) 4117–4136. doi:10.1080/00207543.2021.2020927.

[48] S. Ma, Integrated demand forecasting and inventory replenishment in omni-channel retail with deep learning, International Journal of Production ResearchAdvance online publication (2026). doi:10.1080/00207543.2026.2738209.

[49] Q. Wang, Y. Chen, Y. Mi, C. She, P. Zhao, C. Chen, Contextual distributionally robust optimisation for inventory replenishment optimisation under uncertain demand, International Journal of Production Research 64 (19) (2026) 8393–8411. doi:10.1080/00207543.2026.2653131.

[50] X. Zhang, W. Dai, X. Cai, Inventory replenishment decisions with uncertain price and demand, International Journal of Production Research 62 (7) (2024) 2663–2682. doi:10.1080/00207543.2023.2223701.

[51] M. Gabellini, A. Regattieri, M. Bortolini, M. Ronchi, A novel data-driven approach for integrated inventory replenishment, safety stock and lead time optimisation, International Journal of Production ResearchAdvance online publication (2026). doi:10.1080/00207543.2026.2696440.

[52] G. Schett, C. Ehrig, B. Schwendinger, U. Neumann, N. Beck, F. Stadler, F. Ansari, Integrated demand forecasting and reinforcement learning for order point optimization in inventory planning, International Journal of Production Economics 299 (2026) 110111. doi:10.1016/j.ijpe.2026.110111.

[53] A.-L. Beutel, S. Minner, Safety stock planning under causal demand forecasting, International Journal of Production Economics 140 (2) (2012) 637–645. doi:10.1016/j.ijpe.2011.04.017.

[54] D. K. Barrow, N. Kourentzes, Distributions of forecasting errors of forecast combinations: Implications for inventory management, International Journal of Production Economics 177 (2016) 24–33. doi:10.1016/j.ijpe.2016.03.017.

[55] N. Kourentzes, J. R. Trapero, D. K. Barrow, Optimising forecasting models for inventory planning, International Journal of Production Economics 225 (2020) 107597. doi:10.1016/j.ijpe.2019.107597.

[56] T. Demizu, Y. Fukazawa, H. Morita, Inventory management of new products in retailers using model-based deep reinforcement learning, Expert Systems with Applications 229 (2023) 120256. doi:10.1016/j.eswa.2023.120256.

[57] A. Batsis, S. Samothrakis, Contextual reinforcement learning for supply chain management, Expert Systems with Applications 249 (2024) 123541. doi:10.1016/j.eswa.2024.123541.

[58] A. Kavoosi, R. Tavakkoli-Moghaddam, H. Sajedi, N. Tajik, K. Tafakkori, Dynamic pricing and inventory control of perishable products by a deep reinforcement learning algorithm, Expert Systems with Applications 291 (2025) 128570. doi:10.1016/j.eswa.2025.128570.

[59] A. Kou, Y. Cheng, X. Huang, J. Jin, Dynamic replenishment policy for perishable goods using change point detection-based soft actor-critic reinforcement learning, Expert Systems with Applications 270 (2025) 126556. doi:10.1016/j.eswa.2025.126556.

[60] W. Guo, Z. Ren, W. Du, T. Weng, TFT-MPIR: An end-to-end multi-period inventory replenishment strategy based on temporal fusion transformer, Expert Systems with Applications 261 (2025) 125464. doi:10.1016/j.eswa.2024.125464.

[61] S. Ma, Retail store-SKU level replenishment planning with attribute-space graph recurrent neural networks, Expert Systems with Applications 249 (2024) 123727. doi:10.1016/j.eswa.2024.123727.

[62] T. N. A. Nguyen, T. X. H. Nguyen, N. T. Tran, T. H. Nguyen, P. A. Nguyen, H. A. Vu, Optimizing supply chain operations using advanced time-series mixer models for demand forecasting and inventory under uncertain demand, Expert Systems with Applications 296 (2026) 128955. doi:10.1016/j.eswa.2025.128955.

[63] A. Molina, J. I. Méndez, A. L. Bárcenas-Cortés, A hybrid fuzzy-LLM decision support system for lifecycle-oriented supply-chain strategy, Expert Systems with Applications 334 (2026) 134176. doi:10.1016/j.eswa.2026.134176.

[64] O. Zabraoui, Y. Hmamou, A. Chafi, S. Kammouri Alami, Governed multi-agent reinforcement framework for resilient and sustainable supply-chain decisions: A LangGraph-driven orchestration approach, Expert Systems with Applications 334 (2026) 134471. doi:10.1016/j.eswa.2026.134471.

[65] A. A. Syntetos, J. E. Boylan, On the bias of intermittent demand estimates, International Journal of Production Economics 71 (1–3) (2001) 457–466. doi:10.1016/S0925-5273(00)00143-2.

[66] A. A. Syntetos, J. E. Boylan, On the stock control performance of intermittent demand estimators, International Journal of Production Economics 103 (1) (2006) 36–47. doi:10.1016/j.ijpe.2005.04.004.

[67] N. Kourentzes, Intermittent demand forecasts with neural networks, International Journal of Production Economics 143 (1) (2013) 198–206. doi:10.1016/j.ijpe.2013.01.009.

[68] M. Hasni, M. S. Aguir, M. Z. Babai, Z. Jemai, On the performance of adjusted bootstrapping methods for intermittent demand forecasting, International Journal of Production Economics 216 (2019) 145–153. doi:10.1016/j.ijpe.2019.04.005.

[69] K. Sanguri, S. Patra, S. Punia, Forecast reconciliation in the temporal hierarchy: Special case of intermittent demand with obsolescence, Expert Systems with Applications 218 (2023) 119566. doi:10.1016/j.eswa.2023.119566.

[70] M. Z. Babai, Y. Dallery, S. Boubaker, R. Kalai, A new method to forecast intermittent demand in the presence of inventory obsolescence, International Journal of Production Economics 209 (2019) 30–41. doi:10.1016/j.ijpe.2018.01.026.

[71] I. Svetunkov, J. E. Boylan, iETS: State space model for intermittent demand forecasting, International Journal of Production Economics 265 (2023) 109013. doi:10.1016/j.ijpe.2023.109013.

[72] B. Rostami-Tabar, S. M. Disney, On the order-up-to policy with intermittent integer demand and logically consistent forecasts, International Journal of Production Economics 257 (2023) 108763. doi:10.1016/j.ijpe.2022.108763.

[73] M. Yazdani, T. Aouam, Shipment planning and safety stock placement in maritime supply chains with stochastic demand and transportation times, International Journal of Production Economics 263 (2023) 108952. doi:10.1016/j.ijpe.2023.108952.

[74] T. Zhang, X. Xu, Data-driven lead time forecasting and promising for an online retail platform, International Journal of Production Economics 296 (2026) 109985. doi:10.1016/j.ijpe.2026.109985.

[75] J. D. Croston, Forecasting and stock control for intermittent demands, Operational Research Quarterly 23 (3) (1972) 289–303. doi:10.1057/jors.1972.50.

[76] A. A. Syntetos, J. E. Boylan, The accuracy of intermittent demand estimates, International Journal of Forecasting 21 (2) (2005) 303–314. doi:10.1016/j.ijforecast.2004.10.001.

[77] T. R. Willemain, C. N. Smart, H. F. Schwarz, A new approach to forecasting intermittent demand for service parts inventories, International Journal of Forecasting 20 (3) (2004) 375–387. doi:10.1016/S0169-2070(03)00013-X.

[78] D. Prak, R. Teunter, M. Z. Babai, J. E. Boylan, A. Syntetos, Robust compound Poisson parameter estimation for inventory control, Omega 104 (2021) 102481. doi:10.1016/j.omega.2021.102481.

[79] M. Z. Babai, A. A. Syntetos, R. H. Teunter, Fifty years of inventory research from a forecasting perspective, European Journal of Operational Research 331 (1) (2026) 1–20. doi:10.1016/j.ejor.2025.07.003.

[80] R. L. Goldsmith, A.-L. Sachs, Spare part demand forecasting in every phase: A data pooling approach to the Bass life cycle model, International Journal of Production ResearchAdvance online publication (2025). doi:10.1080/00207543.2025.2607643.

[81] M. Caserta, L. D'Angelo, Optimising two-echelon spare parts inventory: A case study from Amazon's operations, International Journal of Production Research 64 (19) (2026) 8425–8443. doi:10.1080/00207543.2026.2655026.

[82] G. Hadley, T. M. Whitin, Analysis of Inventory Systems, Prentice-Hall, Englewood Cliffs, NJ, 1963.

[83] E. L. Porteus, Foundations of Stochastic Inventory Theory, Stanford University Press, Stanford, CA, 2002. doi:10.1515/9781503619883.

[84] E. A. Silver, D. F. Pyke, D. J. Thomas, Inventory and Production Management in Supply Chains, 4th Edition, CRC Press, Boca Raton, FL, 2017. doi:10.1201/9781315374406.

[85] P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, A. Chadha, A systematic survey of prompt engineering in large language models: Techniques and applications, arXiv preprint arXiv:2402.07927 (2024). doi:10.48550/arXiv.2402.07927.

[86] L. J. Tashman, Out-of-sample tests of forecasting accuracy: an analysis and review, International Journal of Forecasting 16 (4) (2000) 437–450. doi:10.1016/S0169-2070(00)00065-0.

[87] H. Hewamalage, K. Ackermann, C. Bergmeir, Forecast evaluation for data scientists: common pitfalls and best practices, Data Mining and Knowledge Discovery 37 (2) (2023) 788–832. doi:10.1007/s10618-022-00894-5.

[88] S. Watts, G. Shankaranarayanan, A. Even, Data quality assessment in context: A cognitive perspective, Decision Support Systems 48 (1) (2009) 202–211. doi:10.1016/j.dss.2009.07.012.

[89] N. DeHoratius, A. Raman, Inventory record inaccuracy: An empirical analysis, Management Science 54 (4) (2008) 627–641. doi:10.1287/mnsc.1070.0789.

[90] Y. Kang, S. B. Gershwin, Information inaccuracy in inventory systems: stock loss and stockout, IIE transactions 37 (9) (2005) 843–859. doi:10.1080/07408170590969861.

[91] A. J. Mersereau, Information-sensitive replenishment when inventory records are inaccurate, Production and Operations Management 22 (4) (2013) 792–810. doi:10.1111/j.1937-5956.2011.01305.x.

[92] T. J. van Kampen, D. P. van Donk, D.-J. van der Zee, Safety stock or safety lead time: coping with unreliability in demand and supply, International Journal of Production Research 48 (24) (2010) 7463–7481. doi:10.1080/00207540903348346.

[93] M. Thürer, N. O. Fernandes, S. Haeussler, M. Stevenson, Dynamic planned lead times in production planning and control systems: Does the lead time syndrome matter?, International Journal of Production Research 61 (4) (2023) 1268–1282. doi:10.1080/00207543.2022.2034193.

[94] S. Axsäter, Inventory Control, 3rd Edition, Vol. 225 of International Series in Operations Research & Management Science, Springer, Cham, 2015.

[95] P. H. Zipkin, Foundations of Inventory Management, McGraw-Hill, New York, 2000.

[96] M. Bijvank, I. F. A. Vis, Lost-sales inventory theory: A review, European Journal of Operational Research 215 (1) (2011) 1–13. doi:10.1016/j.ejor.2011.02.004.

[97] E. T. Grasso, B. W. Taylor, A simulation-based experimental investigation of supply/timing uncertainty in MRP systems, International Journal of Production Research 22 (3) (1984) 485–497. doi:10.1080/00207548408942468.

[98] H. S. Heese, Minimum order quantity requirements in the presence of supply disruption risk and retail competition, International Journal of Production Research 63 (16) (2025) 6056–6070. doi:10.1080/00207543.2025.2468391.

[99] S. Sengupta, P. Jonsson, H. Dreyer, R. Kaipia, T. Y. Choi, Uncertainty regulation and adaptable supply chain planning, International Journal of Operations & Production Management 45 (11) (2025) 1861–1883. doi:10.1108/ijopm-07-2025-0645.

[100] S. Wasserkrug, L. Boussioux, D. Den Hertog, F. Mirzazadeh, Ş. İ. Birbil, J. Kurtz, D. Maragno, Enhancing decision making through the integration of large language models and operations research optimization, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 39, 2025, pp. 28643–28650. doi:10.1609/aaai.v39i27.35090.